

Isotopic Fractionation

The differences among isotopes of one element are slight. As they have the same number of protons, they will have the same number and arrangement of electrons and, thus, chemically behave interchangeably, forming the same types of chemical bonds as one another. There is one difference, however, which is mass. For light elements, these differences can be significant. For example, most hydrogen has an atomic mass of one (only a single proton, no neutron), but its heavier isotope deuterium also has one neutron, giving it a mass of two—double that of the more common isotope! As atoms become larger, these differences become much slighter, but for light isotopes—including carbon and nitrogen—the differences are meaningful enough to affect biological processes. This is because the mass difference between isotopes affects certain physical properties of the elements, meaning that they can be discriminated against during specific biological processes. This phenomenon in which one isotope is discriminated against, whether in a physical or biological process, is known as **isotopic fractionation**. Isotopic fractionation is often expressed using delta (δ) notation that compares the isotope ratio of a sample with a standard (a universally agreed-upon substance used to set a zero value), in permil units (‰), which are like percentages (%) but 1 in 1,000 instead of 1 in 100. If the sample is identical to the standard, the delta value will be 0‰. Values greater than zero are considered **enriched** in the heavier isotope; those below zero (negative values) are considered **depleted** in the heavier isotope.

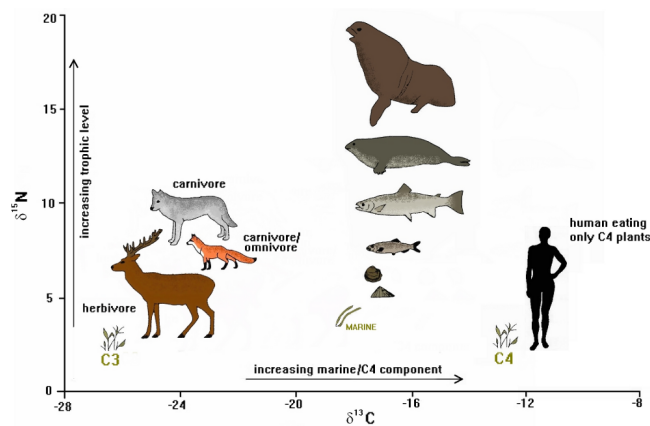
Isotopic fractionation is when different isotopes of the same element behave differently during a physical or chemical process. This can occur as equilibrium or kinetic fractionation; the latter is more relevant for biological processes.

Enriched delta values are those greater than zero. For example, a $\delta^{15}\text{N}$ value of 6.2‰ represents enrichment in the heavy isotope, such as might be found in a predator's skeletal collagen.

Depleted delta values are those below zero (negative). For example, a $\delta^{13}\text{C}$ value of -26.2‰ represents depletion in the heavy isotope, such as might occur during photosynthesis.

$$\delta^{13}\text{C} = \left(\frac{{}^{13}\text{C}_{\text{sample}} / {}^{12}\text{C}_{\text{sample}}}{{}^{13}\text{C}_{\text{standard}} / {}^{12}\text{C}_{\text{standard}}} - 1 \right) \times 1000\text{‰}$$

Above is the equation used to create delta notation for a sample in comparison with a standard, using carbon stable isotopes as an example.



Stable isotope biplot showing $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of different plants and animals. By convention, on such plots, carbon isotope ratios are always given on the horizontal axis, and nitrogen isotope ratios always appear on the vertical axis.

Source: Svyatko, S. *Information about Stable Carbon and Nitrogen Isotope Analysis* (<https://chrono.qub.ac.uk/IRMS/>).

Fractionation of Carbon Isotopes During Photosynthesis

The biochemical processes involved with photosynthesis by plants are complex and beyond the scope of this course (though they are fascinating!). To understand what photosynthesis has to do with isotopic fractionation and the archaeology of foodways, it is sufficient to simply understand that there are two major photosynthetic pathways. These are biochemical processes that plants

have evolved to complete to accomplish photosynthesis. They are known as the C_3 and C_4 pathways; we refer to plants as C_3 or C_4 plants, depending on which pathway they use. There is a third pathway, as well, that is even more complex: crassulacean acid metabolism (usually abbreviated CAM; such plants are called CAM plants). It is much rarer and can be understood as usually similar to C_4 but with flexibility to occasionally look more like C_3 ; we won't focus on it here.

The fundamental biological difference between the C_3 and C_4 pathways is that C_4 uses less water but is somewhat less efficient when water is abundant. The C_3 pathway is the earlier of the two and present by default across the plant kingdom. The C_4 pathway, which evolved multiple times in different lineages of plants, is found generally in hot and/or dry environments. C_4 plants are arid-adapted and mainly include tropical grasses. Important agricultural crops that use the C_4 pathway are maize and East Asian millets (foxtail millet, *Setaria italica*; and broomcorn millet, *Panicum miliaceum*). Most other cultivated crops are C_3 plants, including other cereals (e.g., wheats, barley, oats, rice) and all pulses, as well as nearly all fruits and nuts.

The isotopic difference between C_3 - and C_4 -pathway plants occurs due to the biochemical differences in how air containing carbon dioxide, which is the material converted to sugar during photosynthesis, enters and is chemically bound within a leaf. Glossing over the actual biochemistry of this process, what is important to understand is that both C_3 and C_4 pathways discriminate against the heavier stable isotope of carbon, ^{13}C . This means that $\delta^{13}\text{C}$ values of both C_3 and C_4 plants are negative (depleted). However, C_3 plants discriminate more against the heavier isotope, so they are more depleted (i.e., have larger negative $\delta^{13}\text{C}$ values) than C_4 plants. C_3 plants generally have $\delta^{13}\text{C}$ values between -24‰ and -36‰, while C_4 plants have values between -8‰ and -16‰. Importantly, these values do not overlap. While some isotopic enrichment occurs across trophic levels when plants are eaten by animals (and those animals are eaten by other animals), the significant difference in $\delta^{13}\text{C}$ between C_3 and C_4 plants persists. This means that animal tissues (e.g., collagen or hydroxyapatite extracted from bones of animals) can be distinguished between animals that ate only C_3 plants and those that ate only C_4 plants. If animals eat some of each, their $\delta^{13}\text{C}$ values are intermediate between those of animals eating exclusively C_3 diets and those eating exclusively C_4 plants.

Intermediate values can also be produced by CAM-plant-rich diets, which are rare, or by the consumption of significant quantities of marine foods. This is because the carbon dioxide fixed by marine plants (such as algae and phytoplankton) is enriched relative to atmospheric carbon dioxide used by terrestrial C_3 and C_4 plants, due to a fractionation process that occurs as carbon dioxide dissolves into water. These marine plants are C_3 plants, but marine plant $\delta^{13}\text{C}$ values end up ~8‰ less negative than those of terrestrial C_3 plants, resulting in values intermediate to those of terrestrial C_3 and C_4 plants.

Fractionation of Nitrogen Isotopes by Trophic Level

Nitrogen isotopes are fractionated during a different biochemical process related to the breakdown of nitrogen-containing proteins and their reincorporation into new protein molecules in an animal. This process results in an enrichment in the heavier stable isotope of nitrogen, ^{15}N , relative to the $\delta^{15}\text{N}$ value of the organism eaten. This enrichment amounts to approximately a ~3-4‰ enrichment (values become increasingly large and positive) in $\delta^{15}\text{N}$ relative to the $\delta^{15}\text{N}$ of the

organism eaten. This same enrichment occurs at each step of the food web. Thus, if an herbivore has $\delta^{15}\text{N}$ values of 8-10‰, reflecting an enrichment relative to its plant food sources, a primary carnivore (an animal that eats herbivores) would have $\delta^{15}\text{N}$ values of 11-14‰, and a secondary carnivore (an animal that eats primary carnivores) would have $\delta^{15}\text{N}$ values around 14-17‰. This allows the distinction of the trophic level of an animal based on its $\delta^{15}\text{N}$ values. Humans who are vegans can be distinguished from those who consume significant quantities of meat on this basis.

Nitrogen stable-isotope values are most useful, however, in distinguishing aquatic (especially marine, but also freshwater) diets from terrestrial ones. Humans infrequently eat carnivores; with most of human animal diet comes from the tissues (meat or milk) of herbivores, such as chickens, cattle, sheep, deer, and rabbits. As a result, humans eating a diet that includes significant quantities of meat or other animal products are expected to resemble those of primary carnivores, with $\delta^{15}\text{N}$ values up to 11-14‰. Aquatic—especially marine—ecosystems, however, contain many more layers to a food web. For example, phytoplankton are eaten by zooplankton, which are eaten by filter-feeding animals; which are then eaten by carnivorous animals, such as fish, which are eaten by larger fish, and yet larger fish; and these are perhaps eaten by top predators, such as sharks or marine mammals (e.g., toothed whales). If humans eat those top predators—which are often preferred prey since they are such large animals—they may be eating five or more levels up a food web. Since $\delta^{15}\text{N}$ enrichment occurs at each trophic step, the $\delta^{15}\text{N}$ values of humans who eat large quantities of marine animals can be as high as 20‰.